CampusNet+ System Specification

India's Premier National Academic, Innovation & Research Network
A unified digital ecosystem connecting higher education institutions, faculty guides, PhD scholars, and students across all 28 States and 8 Union Territories.

> **Executive Mandate:** CampusNet+ serves as an end-to-end national digital backbone for collegiate project incubation, Smart India Hackathons (SIH), geofenced attendance tracking, tamper-proof credential issuance, and inter-university research publication discovery.

1. The Problem Space in Indian Higher Education

Innovation in Indian higher education has historically been hindered by systemic fragmentation:

- **Institutional Isolation:** Talented students in tier-2/tier-3 engineering colleges cannot readily partner with specialists or find research guides at premier institutes (IITs, NITs, IISc).
- **Hackathon Proxy Attendance:** On-campus and inter-college hackathons struggle with manual paperwork, duplicate registration, and proxy check-ins.
- **Unverifiable Paper Certificates:** Physical and standard PDF certificates of merit or participation lack machine-readable, cryptographic authentication.
- **Underutilized Academic Research:** Capstone projects and student publications frequently go unindexed, preventing commercial incubation or cross-disciplinary collaboration.

2. Core Platform Statistics & Pan-India Footprint

Ecosystem Entity	Production Metric	Primary Administrative Function
Student Innovators	10,000+ Enrolled	Team formation, milestone execution, 1-page Innovation Transcript.

Ecosystem Entity	Production Metric	Primary Administrative Function
Faculty Mentors	850+ Verified	Vidwan profile integration, project guidance, milestone review.
PhD Research Scholars	520+ Active	Peer reviews, research paper registry, open lab collaborations.
Colleges & Universities	450+ Campuses	AISHE accredited institutions across all 28 Indian States & 8 UTs.
Hackathons & Fests	1,200+ Hosted	Nodal GPS/QR check-in terminals, weighted jury rubrics, awards.

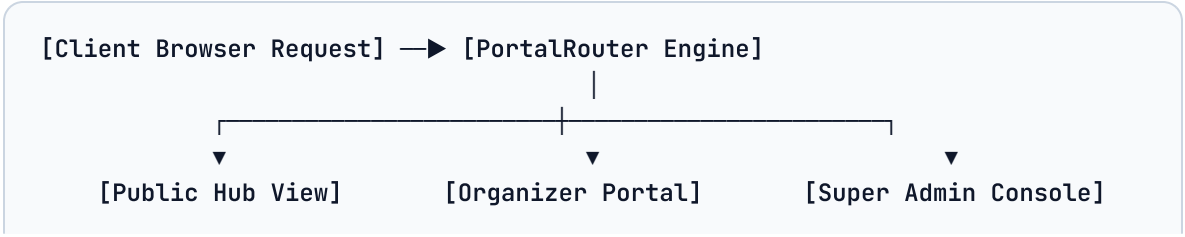
System Architecture & Multi-Portal Engine

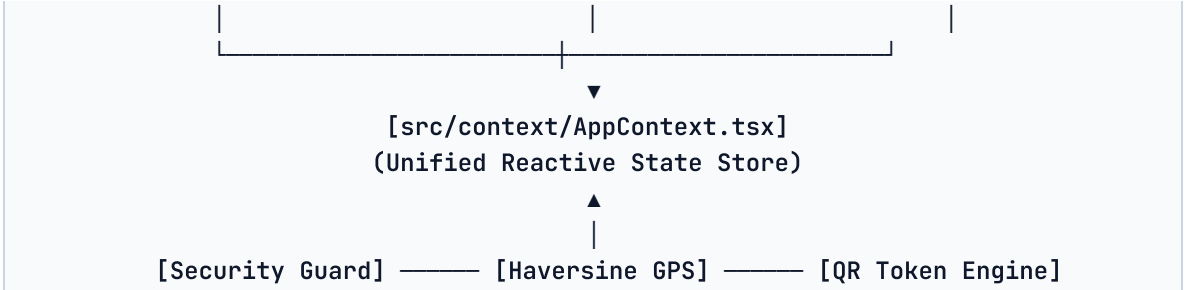
CampusNet+ utilizes a 3-tier enterprise architecture separating public community engagement from event administration and national governance.

1. Dedicated 3-Tier Portal Routing Matrix

Portal Instance	Production Domain	Local Development Route	Target Audience & Access
🌐 **Public CampusNet**	`campusnet.in`	`http://localhost:3000/`	Students, Faculty Mentors, Researchers, Public Visitors.
🏢 **Organizer Portal**	`organizer.campusnet.in`	`http://localhost:3000/?portal=organizer`	University Event Chairs, Department HODs, Jury Panels.
🛡️ **Super Admin Console**	`admin.campusnet.in`	`http://localhost:3000/?portal=admin`	National Governance, Accreditation Officers (MFA Protected).

2. Unified Reactive Core Data Flow





3. Security Isolation Policy

The Public Portal's top navigation bar (`Navbar.tsx`) and `Footer.tsx` contain **zero exposed links** to the Organizer Portal or Super Admin Console. Access to administrative tiers is controlled via subdomain DNS routing in production and secure query token fallbacks in development environments.

Technology Stack & Codebase Map

Modern, ultra-fast frontend and lightweight production server engineered for zero latency and continuous deployment.

1. Technology Stack Specifications

Component Layer	Technology & Version	Architectural Role
Core Frontend	React 19.0.0	Component modularity, concurrent rendering, hooks architecture.
Language	TypeScript 5.7.3	Strict typing, data integrity, compile-time contract enforcement.
Build Tool	Vite 6.2.0	Sub-second Hot Module Replacement (HMR) and optimized Rollup chunks.
Styling	Tailwind CSS 3.4.17	Curated academic theme (Navy Blue, Crimson Red, Slate, Warm White).
Client Routing	React Router DOM 7.18.2	Dynamic parameter handling, smooth scroll restoration, nested layouts.
Production Server	Node.js HTTP (server.js)	Native SPA fallback server with immutable asset caching policies.

2. Directory Structure Overview

```
c:\projects\CampusNet\
├── server.js               # Node.js production SPA HTTP fallback server
├── vite.config.ts          # Vite build configuration with chunking rules
├── src/
│   ├── App.tsx             # Root application shell & modal overlays
│   ├── context/            # AppContext.tsx (2,395-line unified reactive store)
│   ├── types/              # Master TypeScript domain models & interfaces
│   ├── data/               # Indian academic seed database (colleges, events, c
│   ├── utils/              # 28 States + 8 UTs location data, validation utilit
```

```
│   └── components/
│       ├── admin/          # National Super Admin Console
│       ├── organizer/      # Institutional Event Organizer Portal
│       ├── team/           # Private 6-Member Workspace & Kanban
│       ├── certificates/   # Tamper-proof certificate generator & QR verify
│       └── dashboard/      # Student, Mentor & Researcher personal views
```

Domain Models & TypeScript Entities

Enterprise-grade data structures declared in `src/types/index.ts` enforcing strict consistency across all tiers.

1. User & Identity Entities

```
export interface User {
  id: string;
  name: string;
  email: string;
  mobile: string;
  studentId?: string;
  institution: string;
  department: string;
  course: string;
  year: string;
  verifiedStudent: boolean;
  innovationScore: number;
  skills: string[];
  badges: string[];
  role: 'student' | 'mentor' | 'researcher' | 'organizer' | 'superadmin';
}
```

2. Event & Geofencing Model

```
export interface EventItem {
  id: string;
  code?: string;                // e.g. KEC-AI26-HACK
  title: string;
  organizer: string;
  eventType: 'Hackathon' | 'Ideathon' | 'Government Challenge';
  state: string;
  city: string;
  mode: 'Offline' | 'Online' | 'Hybrid';
  attendanceWindow: {
    start: string;
    end: string;
    targetLat: number;         // GPS Latitude
    targetLng: number;         // GPS Longitude
```

```
    allowedRadiusMeters: number; // e.g. 500m
  };
  prizes: EventPrize[];
  status: EventStatus;
}
```

3. Digital Credential & Verification Model

```
export interface Certificate {
  id: string;
  certificateNumber: string;    // e.g. CN-KEC-AI26-PART-000101
  recipientName: string;
  recipientRole: 'Participant' | 'Winner' | 'Runner-Up' | 'Mentor';
  eventTitle: string;
  eventOrganizer: string;
  issueDate: string;
  qrCodeData: string;
  status: 'valid' | 'revoked';
  revocationReason?: string;
}
```

Public Platform Modules & Discovery Engine

National hub for cross-college project incubation, faculty mentor matching, and official competency transcripts.

1. Universal Federated Search

The search engine at `FederatedSearchDropdown.tsx` enables students and faculty to query across five domains simultaneously with keyboard shortcuts (`ArrowUp` , `ArrowDown` , `Enter`):

- **Students:** Search by university roll number, skills (e.g., ROS2, PyTorch), or state.
- **Projects:** Search by problem statement, required open roles, or domain.
- **Mentors:** Search by research focus, h-index, or Vidwan scholar profile.
- **Events:** Filter national hackathons by prize pool, offline mode, and deadline.

2. Faculty Mentorship Matchmaking Heuristic

The algorithm calculates a 0–100% compatibility score evaluating domain alignment, technology overlap, mentor research publications, and current active team load:

```
CompatibilityScore = (0.35 × DomainMatch) + (0.30 × TechMatch) + (0.20 × ResearchMatch) + (0.15 × SlotAvailability)
```

3. Official 1-Page AI Innovation Transcript

Accessible via `/portfolio` , students can preview and print an official single-page summary (`InnovationTranscriptModal.tsx`) designed for job placements and higher studies:

Section	Verified Data Displayed
Student Header	AISHE college name, degree, department, bonafide verification badge.
Innovation Index	Composite score (0–1000) derived from completed milestones and podium finishes.
Verified Credentials	Serial-numbered hackathon certificates and faculty mentorship citations.
QR Authentication	Dynamic QR code linking directly to the live digital verification page.

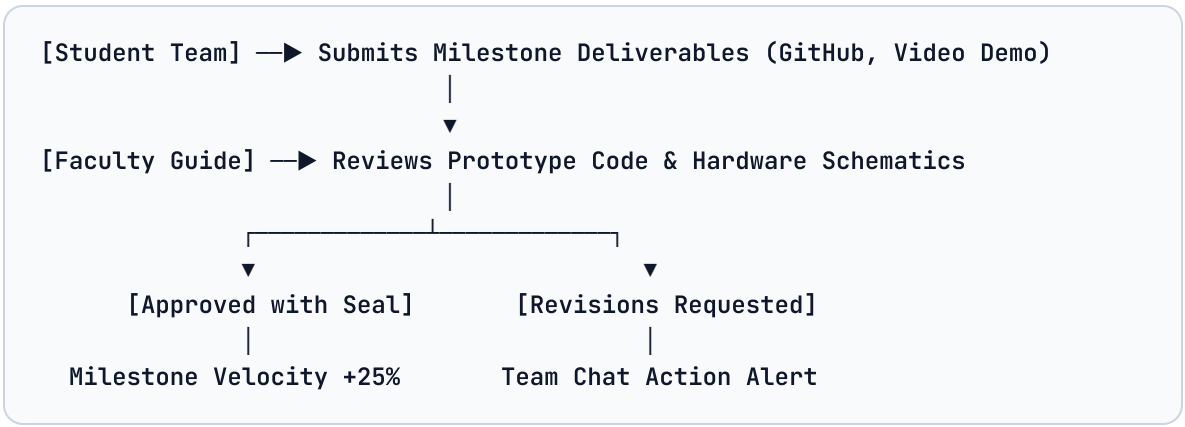
Team Collaboration Workspace & Real-Time Studio

Private 6-member Team Studio (`/workspace/:teamId`) providing real-time tools for multi-disciplinary project execution.

1. Collaborative Studio Architecture

Module Name	Component Path	Core Capabilities
Kanban Board	`KanbanBoard.tsx`	3 swimlanes (Todo, In Progress, Done), priority flags, milestone sync.
Team Chat Stream	`TeamChat.tsx`	Persistent chronological messaging, code blocks, schematic uploads.
WebRTC Video Console	`VideoCallModal.tsx`	Simulated video/mic toggles, screen share, active speaker detector.
Mentor Guidance	`MentorGuidanceTab.tsx`	Milestone reviews, formal sign-offs, completion certificate issuance.

2. Project Milestone Review Flow



Institutional Event Organizer Portal

Dedicated operations console accessible via `?portal=organizer` for colleges hosting state and national technical hackathons.

1. 30+ Field Event Creation Wizard

The wizard (`EventCreationWizard.tsx`) guides event chairs through four structured stages:

- **Step 1: Event & Coordinator Identity:** Title, system code, event tracks, prizes, and coordinator direct contacts.
- **Step 2: Geofencing Coordinates:** State, district, venue address, target latitude/longitude, and permissible check-in radius (meters).
- **Step 3: Eligibility & Rules:** Eligible departments, batch years, team sizes, and schedule milestones.
- **Step 4: Formal College Sanctions:** Mandatory upload of dean approval letters (.pdf, .png, .jpg up to 10 MB).

2. Dual-Mode Attendance Terminal

The check-in scanner terminal (`AttendanceQRScannerView.tsx`) eliminates attendance fraud during on-campus hackathons:

Check-In Mode	Verification Process	Fraud Prevention Mechanism
Optical QR Scanner	Reads participant dynamic QR badge via device camera.	Rejects duplicate scans; displays first check-in timestamp.
Manual Override	Search by University Registration Number or Student ID.	Logs coordinator name, timestamp, and verification reason.

3. Weighted Jury Judging & Leaderboard

External judges authenticate using designated keys (e.g. `JUDGE-KEC-RAO-99`) to grade submissions across Innovation (30%), Technical Complexity (30%), Practical Viability (25%), and Presentation (15%). Total weighted scores dynamically populate the Winner Selection console.

National Super Admin & Governance Console

National accreditation and compliance terminal accessible via `?portal=admin` protected by hardware MFA tokens.

1. National Event Moderation Queue

Located at `SuperAdminEventModeration.tsx`, this queue oversees all collegiate proposals across India:

- Inspects college sanction letters and faculty coordinator identities.
- Super Admins can **Approve & Publish** to the public directory, **Request Changes** with inline reviewer notes, or **Reject with Written Cause**.

2. College Accreditation & User Moderation

Module Name	Governance Responsibility
Institution Manager	Verifies AISHE codes, official .ac.in / .edu.in domain patterns, and grants verified green badges.
User Moderator	Inspects student identity cards, assigns bonafide badges, or suspends violators.
Audit Trail Log	Records immutable audit entries (Actor ID, Target Entity, IP, Action, Timestamp).

3. Certificate Fraud Auditor & Instant Revocation

The national certificate index (`SuperAdminCertificateAuditor.tsx`) enables instant revocation of falsified or disputed credentials. When revoked, the public verification page immediately flags the certificate with a bold red warning displaying the revocation timestamp and auditor justification.

Security Engineering & Anti-Fraud Mechanisms

Mathematical and cryptographic protocols preventing proxy attendance, credential falsification, and injection attacks.

1. Geofencing Mathematics: Haversine Distance Formula

To verify that participants are physically present at the venue before marking attendance, the system evaluates the great-circle distance between device GPS coordinates and the venue anchor:

$$a = \sin^2(\Delta\varphi/2) + \cos(\varphi_1) \times \cos(\varphi_2) \times \sin^2(\Delta\lambda/2)$$
$$c = 2 \times \operatorname{atan2}(\sqrt{a}, \sqrt{(1-a)})$$
$$\text{distance} = \text{EarthRadius}\ (6{,}371{,}000\text{m}) \times c$$

If `distance > allowedRadiusMeters`, the check-in is rejected, preventing remote proxy fraud.

2. Cryptographic QR Verification & Public Lookup

Every certificate generated on CampusNet+ is assigned a unique, collision-resistant identifier and QR payload:

```
Certificate ID Format: CN-[INSTITUTION]-[EVENT_CODE]-[ROLE]-[SERIAL_HASH]
Example: CN-KEC-AI26-PART-000101

QR Payload URL: https://campusnet.in/verify/CN-KEC-AI26-PART-000101
```

3. Strict URL Allowlists & File Sanitization (validation.ts)

- **Domain Allowlists:** External profile URLs are strictly validated against `github.com`, `linkedin.com`, `x.com`, `leetcode.com`, and `vidwan.inflibnet.ac.in`.

- **File Upload Defense:** 10 MB size limit, PDF/image MIME enforcement, path traversal stripping (`../`), and blocking of executable formats (`.exe` , `.sh` , `.php` , `.js`).

Deployment Guide & Strategic Roadmap

Production runtime instructions, master demo credentials, and upcoming technical milestones.

1. Local Setup & Execution

```
# 1. Clone repository
git clone https://github.com/mogliesworachary-droid/campusnet.git
cd campusnet

# 2. Install dependencies
npm install

# 3. Start local development server
npm run dev -- --host
# Access Public Hub: http://localhost:3000/
# Access Organizer:  http://localhost:3000/?portal=organizer
# Access Admin:      http://localhost:3000/?portal=admin

# 4. Compile Production Bundle
npm run build && npm start
```

2. Master System Demo Credentials

Role	Email / Access Key	Password / Token
🛡️ **Super Admin**	`superadmin@campusnet-demo.in`	`Admin@CampusNet2026` (MFA: `849201`)
🏢 **Organizer (KEC)**	`organizer.demo@campusnet-demo.in`	`CampusNet@Demo2026`
⚖️ **Jury Judge 1**	Key: `JUDGE-KEC-RAO-99`	Assigned Event: `ev-kec-001`

3. Strategic Technical Roadmap

1. **Academic Bank of Credits (ABC) & DigiLocker:** Synchronizing verified hackathon finishes directly with the Ministry of Education's National Academic Depository.
2. **Real-Time WebSocket Canvas:** Upgrading team workspaces with live multi-user whiteboards and cursor synchronization.
3. **Polygon L2 Smart Contract Anchoring:** Merkle-root notarization of issued certificates for global cryptographic proof of academic awards.

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